CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · PSYCHIATRY & MENTAL HEALTH PHASE 2,3 · STARTS 2015–2024

# Monoamine reuptake: 10 of 20 completed trials missed their primary endpoint

**10 completed trials across 8 sponsors and 8 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 10 met it.** Only 3 of 67 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

**10 of 20**
completed trials with a readable primary result missed it · 10 met it

**8 / 8**
sponsors / molecules behind the misses

**4.5%**
stopped early: 3 of 67 closed trials (95% CI 1.5%–12.4%)

**74.4%**
of 90 trials have closed · 23 still open or unresolved

**10 of 20** **Completed and missed the primary endpoint.** Of 57 completed trials in this segment, 20 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 10 missed and 10 met. Counted separately and never in the rate above; a floor, not a rate. 12 of the 54 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## The completed trials that missed

| Trial | Ph | Sponsor | Experimental drugs | Started | What the posted result says |
|---|---|---|---|---|---|
| NCT04885257 | 2 | VA Office of Research and Development | METHYLPHENIDATE | 2022-01 | Change in Modified Rankin Scale at 12 Weeks — p 0.896, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Method: Wilcoxon (Mann-Whitney). |
| NCT04437485 | 2 | Indiana University | BUPROPION | 2020-10 | Hemoglobin A1c at 6 Months — p 0.64, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Method: ANCOVA. |
| NCT04345471 | 3 | Mochida Pharmaceutical Company, Ltd. | DESVENLAFAXINE | 2020-05 | Changes in Total MADRS Score From the Baseline to Week 8 Visit During the Treatment Period — p 0.509 against the sponsor's stated threshold of 0.05: not significant. LS mean difference -0.6 (95% CI -2.5 to 1.2). Method: MMRM. |
| NCT03569475 | 3 | AbbVie | LEVOMILNACIPRAN | 2018-07 | Change From Baseline in Children's Depression Rating Scale-Revised (CDRS-R) — p 0.2215, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Least Square Mean Difference -1.7 (95% CI -4.49 to 1.04). Method: Mixed Model for Repeated Measures. |
| NCT03315793 | 3 | Shionogi | DULOXETINE | 2017-12 | Change From Baseline on the Children's Depression Rating Scale-Revised (CDRS-R) Total Score — p 0.5587, no threshold stated; the sponsor posted a 98% interval, i.e. a design tested at two-sided 0.02: not significant. Least squares mean difference 1.39 (98% CI -3.3 to 6.08). Method: mixed-effects model repeated measures. |
| NCT02709746 | 3 | H. Lundbeck A/S | VORTIOXETINE | 2016-05 | Change in Children Depression Rating Scale - Revised (CDRS-R) Total Score After Treatment — p 0.4702, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Mean Difference (Final Values) 1.13 (95% CI -1.94 to 4.2). Method: Mixed Model Repeated Analysis. |

The 6 most recently started of 10 completed trials that missed. The diligence report lists every one, with the numbers or the sentence each verdict was read from.

## How this compares

Monoamine reuptake



4.5%

3/67 closed

all psychiatry & mental health trials with a resolved drug target



3.6%

15/411 closed

All psychiatry & mental health Phase 2,3



2.9%

30/1024 closed

**What is in this class:** 119 distinct experimental drugs across 90 trials (small molecule 90, unknown 3). Most tested: LUMATEPERONE (17), VORTIOXETINE (14), MIDOMAFETAMINE (11), DULOXETINE (10), FLUOXETINE (6). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 3 stops came from 2 sponsor-asset programmes across 2 sponsors; the largest (H. Lundbeck A/S) contributed 2; removing that programme's trials from both sides leaves 1.7%. Screened alongside 12 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. One further closed trial here was terminated with no cause recorded in the registry; if it were biological the rate would be 6.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2015–2017 | 1 | 28 | 3.6% | 0.6%–17.7% |
| 2018–2020 | 0 | 17 | 0.0% | 0.0%–18.4% |
| 2021–2024 | 2 | 22 | 9.1% | 2.5%–27.8% |

## Where the stops sit

Efficacy / futility: 2, Safety: 1.
Sponsors with most stops: H. Lundbeck A/S (2), Scripps Health (1).

## The stopped trials

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT05611749 | 2 | Scripps Health | DULOXETINE | 2024-03 | Safety | Enrollment was extremely low and there were too many adverse events. |
| NCT05014919 | 3 | H. Lundbeck A/S | VORTIOXETINE | 2022-03 | Efficacy | The study was terminated based on new efficacy data from another study. |
| NCT02871297 | 3 | H. Lundbeck A/S | VORTIOXETINE | 2022-03 | Efficacy | The study was terminated based on new efficacy data from another study. |

**Method**
Denominator: ClinicalTrials.gov interventional Phase 2,3 psychiatry & mental health trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 44% of industry psychiatry & mental health trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 74.4% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.